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The application of solar-to-thermal conversion phase change material in novel solar water heating system

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ABSTRACT

Solar energy has been widely used in water heating systems. Latent thermal storage technology using phase change material (PCM) has also attracted great attention in energy storage field. In this study, two kinds of composite PCM- sodium acetate trihydrate/expanded graphite (SAT/EG) and sodium acetate trihydrate/graphene oxide/expanded graphite (SAT/GO/EG) were prepared using a physical hybrid method. The melting temperature and latent heat of SAT/EG were 57.9 °C and 218.1 J/g, respectively. GO showed a negligible influence on the thermal thermophysical characteristics of composite PCM. Exposed in solar radiation, two PCMs can experience phase change almost at same rate. The leakage tests were conducted on these PCMs with the purpose of confirming the density of PCMs. The highest densities at which the PCMs can keep good shape stabilities were 0.8 g/cm^3 for SAT/EG and 1.0 g/cm^3 for SAT/GO/EG, which indicates that traces of GO (less than 0.1 wt%) can improve the shape stability of compressed PCMs. A solar water heating system equipped with a new type of collector was constructed by employing SAT/GO/EG (1.0 g/cm^3, 1628 g) as the solar-absorbing substance and thermal storage medium simultaneously. Under real solar radiation, the SAT/GO/EG can finish phase change and reach the temperature above 70 °C from 9:00 AM to 2:00 PM on a typical day in Zhuhai, China. The solar-to-thermal conversion efficiency of the system was calculated to be 54.5%. These results illustrate that SAT/GO/EG possesses excellent solar-absorbing performance and shows great promise in designing novel solar water heating system.

1. Introduction

Solar energy has been widely used in the area of industry, building and other thermal supply system due to its cleanliness and abundance (Qi et al., 2017; Khan et al., 2018). Among these applications, solar water heating system is the most popular especially in the area of plateau regions and low latitudes (Wu et al., 2018; Mandal et al., 2019). A typical solar water heating system consists of four components: solar collector, water tank, pipelines and control system. Collector is the most crucial because it functions as heat source in charge of transferring solar radiation into thermal (Kalogirou, 2004). This thermal energy is commonly reserved in water by the way of sensible heat. However, sensible heat storage suffers from lower energy density. Compared the sensible thermal storage, latent heat storage using phase change material (PCM) has the advantages of high storage capacity and narrow temperature variation (Islam and Morimoto, 2018; Carmona and Palacio, 2019). Hence, a relatively smaller volume of PCM is sufficient to store the required amount of heat (Kee et al., 2018). That is to say, replacing water by PCM can greatly reduce the volume of thermal storage unit (e.g. water tank). Therefore, combining phase change heat storage technology and traditional solar water heating system has attracted great attention. Li et al. proposed a collector/storage system using erythritol as PCM and simulated its heat transfer characteristics (Li et al., 2018). The results showed that the system with composite PCM had a good thermal storage performance, and the daily average storage efficiency reached 39.98%. Al-Hinti et al. refitted a conventional solar water heating system by installing two levels of PCM in a water tank. It was found that the refitted system can result in a 13–14 °C advantage in the water temperature over extended periods (Al-Hinti et al., 2010). Wu et al. constructed a novel solar water heating system capable of reducing the impact of solar radiation intensity fluctuations by utilizing PCM and oscillating heat pipe. It was proved that the collecting efficiency fluctuation with PCM is over 30% less than that without PCM (Wu et al., 2018).

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In above cases, the PCMs just function as the thermal storage media while the collectors are responsible for the thermal generation (Shirinbakhsh et al., 2018). The thermal produced by the collectors are transported to the PCMs through water or other fluid, in which two heat exchange processes are carried out. One occurs between collector and water, and other occurs between water and PCM. These two processes inevitably give rise to decreased energy efficiency and anabatic thermal loss. In addition, the solar heating systems using water as heat transferring medium or thermal storage medium are not applicable to severe cold areas due to freeze phenomenon. Therefore, other application mode of PCM in solar heating system should be explored.

That PCM can be endowed with excellent solar-to-thermal conversion performance has been verified by many researches (Huang et al., 2014; Wang et al., 2012, 2013). In our previous work, several solar-to-thermal conversion PCMs were also prepared, and their conversion efficiency was higher than 80% (Xiao et al., 2018; Xiao et al., 2018). It can be expected that employing these materials as the solar-absorbing substance and heat storage media simultaneously can construct a novel solar collector as well as a water heating system. In this novel system, the function of traditional collector such as evacuated tube collector and flat plate collector is realized by the solar-to-thermal conversion PCM. Consequently, in new system, two heat exchanges mentioned above would be avoided.

$NaAc{\cdot}3H_2O$ (SAT) is a desirable hydrated salt PCM using in the water heating system owing to its merits of high phase change latent heat (278 J/g) and suitable phase change temperature (58 °C) (Shin et al., 2015; Hu et al., 2011). However, pure SAT encounters low thermal conductivity and leakage in the state of liquid, which make it not suitable for the application in water heating system (Xiao et al., 2019). Our previous research indicated that combining SAT and expanded graphite (EG) to prepare SAT/EG composite PCM is a feasible method to overcome such disadvantages. More importantly, this composite PCM is characterized by superior solar-to-thermal conversion performance (Xiao et al., 2018). As a result, it has great promise to build the novel solar water heating system.

In the current work, composite PCMs with and without graphene oxide (GO) were prepared. Their phase change properties, thermal stabilities and thermal conductivities were examined using a differential scanning calorimeter (DSC), a thermal gravimetric analyzer (TG) and thermal constant analyzer. Solar-absorbing performance of these PCMs was compared by placing the PCMs under the sun. In order to improve bulk heat storage density but maintain the shape stability of composite PCMs, leakage tests were conducted on these PCMs with series of compressed densities. Afterwards, a novel solar water heating system employing SAT/GO/EG at proper density as solar-absorbing substance and heat storage media simultaneously was constructed. The solar-to-thermal conversion efficiency of the system was investigated under a real environment. Finally, the application effect of novel system was observed through a heat exchange process between PCM and 10 L water.

2. Experimental sections

2.1. Materials

Anhydrous sodium acetate (NaAc, AR) and Disodium hydrogen phosphate (Na_2HPO_4, AR) were purchased from the Shanghai Macklin Biochemical Co., Ltd. (Shanghai, China). Graphene oxide dispersion (GO, 1 wt%) was supplied by the Chengdu Organic Chemicals Co., Ltd. (Chengdu, China). Expanded graphite (EG, 80 mesh) was provided by the Qingdao Graphite Co., Ltd. (Qingdao, China). Before use, it was dried at 120 °C for 5 h to eliminate absorbed water. Deionized water was made in the laboratory.

2.2. Sample preparations

2.2.1. Preparation of SAT/EG

NaAc and deionized water at a molar ratio of 1:3 were thoroughly mixed at the temperature of 90 °C until the NaAc was totally dissolved. 2 wt% of Na_2HPO_4 was then added into the above mixture. After continuous heating for another 1 h, the hydrated salt composite containing $NaAc{\cdot}3H_2O$ and nucleating was well synthesized and labeled as SAT for convenience. The obtained SAT and EG were further blended at a mass ratio of 8:2 to prepared SAT/EG composite. It can be acquired after appropriate stirring for 30 min at the temperature of 90 °C.

2.2.2. Preparation of SAT/EG/GO

1 wt% GO solution was first diluted using deionized water. NaAc was then dissolved in the diluent GO dispersion at 90 °C to form a $NaAc/GO/H_2O$ mixture in which the molar ratio of NaAc to H_2O was 1:3 and the mass fraction of GO was 0.1%. Na_2HPO_4 (2 wt%) employed as nucleating agent was added into above salt solution. The mixture was heated at 90 °C and stirred for another 1 h to obtain SAT/GO composite. Similar with the SAT/EG, the SAT/GO/EG was also prepared by blending SAT/GO and EG at the same ratio.

2.3. Characterizations

The phase change properties of composite PCMs were investigated using a differential scanning calorimeter (DSC, NETZSCH 214 Polyma, Germany). The sample was sealed in an aluminium crucible and heated from 0 °C to 75 °C. A thermal gravimetric analyzer (TG, NETZSCH F3 209, Germany) was carried out to analyze the thermal degradation at the temperature range 30–700 °C. Thermal conductivities of SAT/EG and SAT/EG/GO were measured using a thermal constant analyzer (Hot Disk TPS-2500, Sweden). The test samples were prepared by compressing loose SAT/EG and SAT/EG/GO into cylindrical blocks (4.0 cm in diameter and 1.5 cm in height) with a series of densities (0.4 g/cm^3, 0.6 g/cm^3, 0.8 g/cm^3, 1.0 g/cm^3, 1.2 g/cm^3). A 7577 probe was switched between two same samples. These compressed samples with various densities were also used to evaluate the shape stability in order to determine the most suitable density of PCMs used in designing water heating system. The temperature was measured and recorded using a data logger (Agilent 34970A, USA) with K-type thermocouples with an accuracy of 0.5 °C (Figs. S1–S3 in Supporting Information).

2.4. Construction of novel solar water heating system

2.4.1. Structure the system

As shown in Fig. 1(a) and (b), the solar water heating system basically contains three components: a pump, a water tank and a novel collector (Fig. 1(d)). The collector is also composed of three parts: PCMs, an aluminium alloy framework used to hold PCM and a cold plate where the heat exchange between PCMs and water takes place (Fig. 1(c)). Both aluminium alloy framework and cold plate are made from aluminium alloy (6061). For convenience, they are collectively named as PCM chamber. There are four parallel channels running through the cold plate. Cold water flows through the cold plate and takes thermal energy away from the collector. The PCM is packed into the four cavities of aluminium alloy framework by cold compression, as displayed from Fig. 1(d)–(e). Each cavity is 185 mm in length, 110 mm in width and 20 mm in height. Therefore, the upper surface area and volume of total PCM are 814 cm^2 and 1628 cm^3, respectively. After loaded with PCM, the PCM chamber is covered by a piece of acrylic board (3 mm in thicknesses, Dongguan Yaxin Electronic Technology Co., Ltd.) to separate the PCM from external environment.

2.4.2. Performance evaluation

The collector was horizontally placed on the rooftop and exposed in the sunlight. Under solar radiation, the PCM can absorb radiation

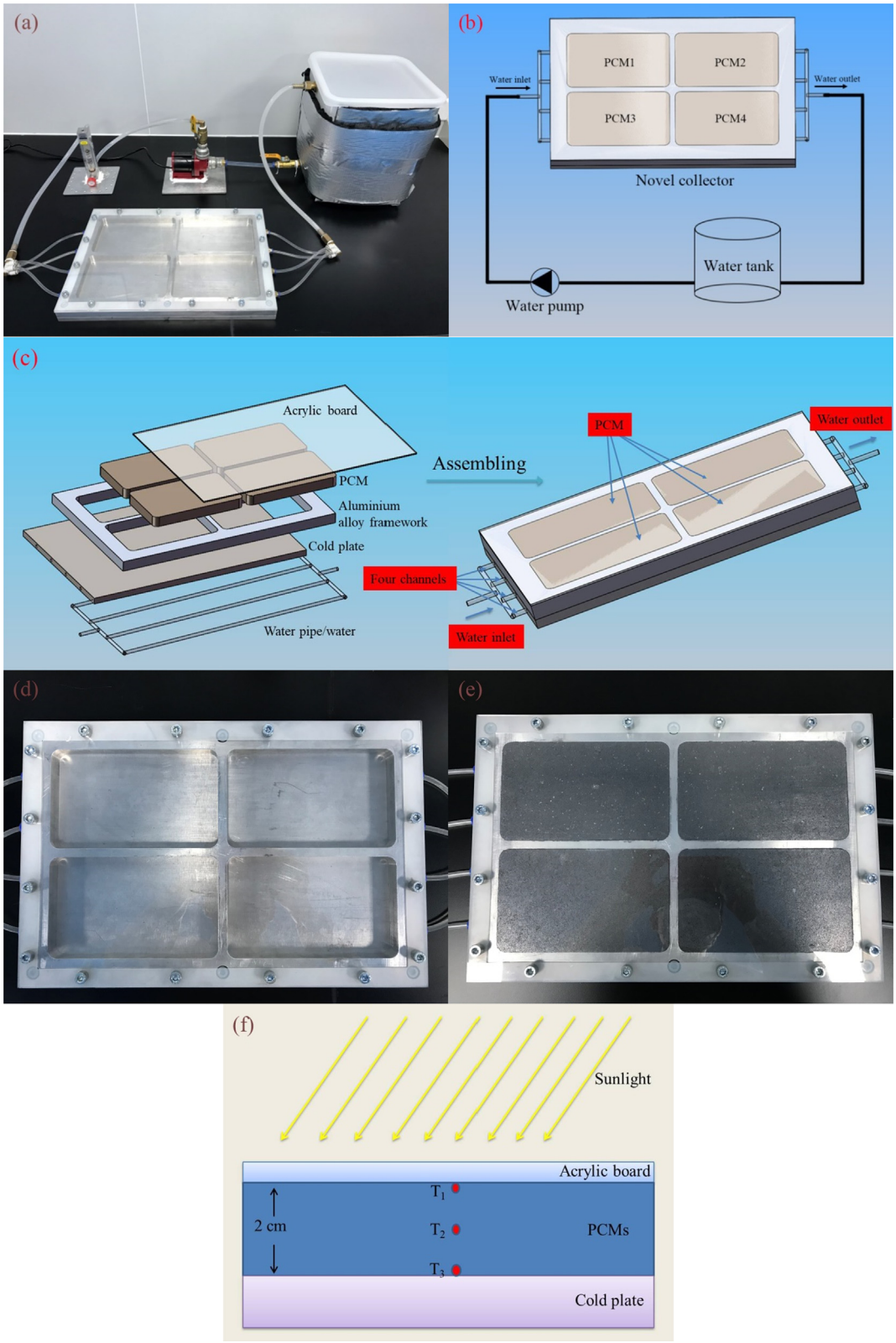


Fig. 1. Photograph (a) and structure diagram (b) of novel solar heating water system, the schematic diagram of novel collector (c), the photograph of collector without (d) and with (e) the PCM, positions of temperature measuring points (f).

energy and transfer solar energy to thermal energy. Therefore, the temperature of the PCM gradually increased with the increasing time. Phase change occurred once the temperature reached the melting point of the PCM. After the phase change finished, the temperature continued to increase. Three temperature measuring points marked as T_1, T_2 and T_3 were arranged at the surface, middle and bottom of the PCM, as displayed in Fig. 1(f). The average temperature of these three points was regarded as the PCM temperature. The solar radiation measurement system consists of three parts: a total solar radiation sensor (NHZF51, Wuhan Zhongke Nenghui Technology Development Co., Ltd.,

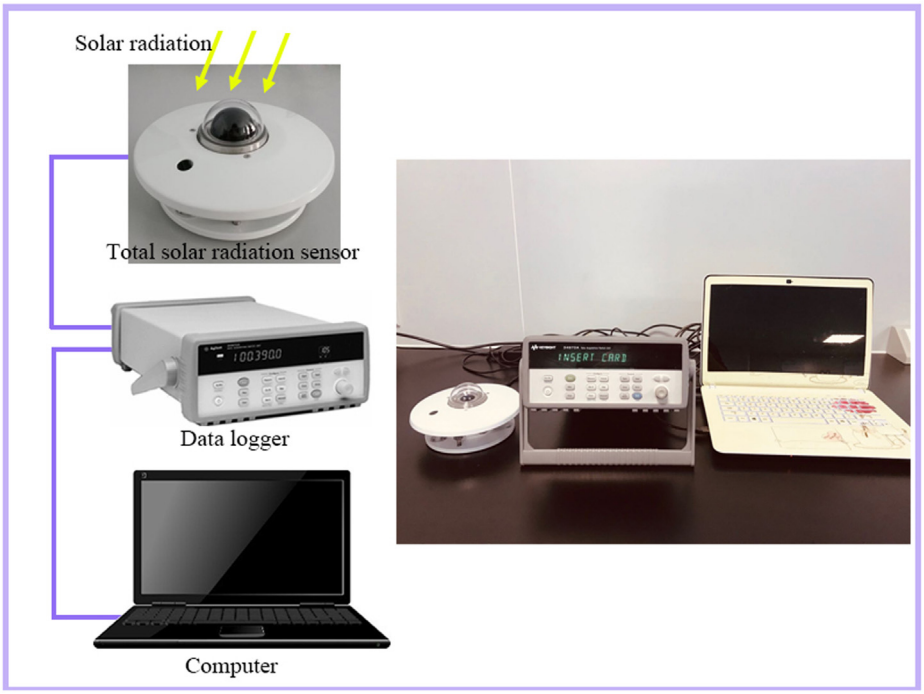


Fig. 2. The experimental setup of solar radiation measurement system.

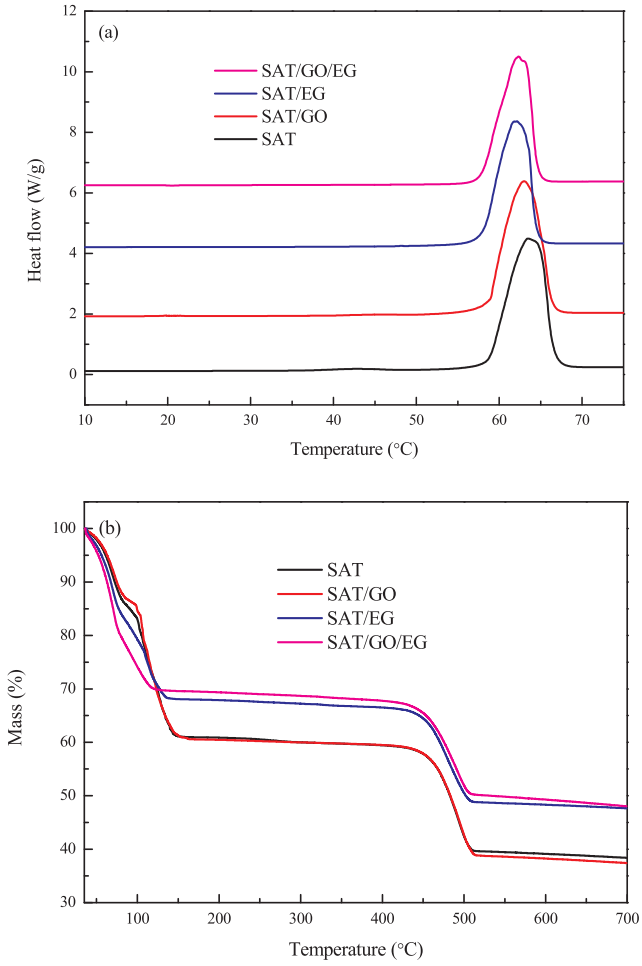


Fig. 3. DSC (a) and TG (b) curves of composite PCMs.

China), a data logger (Agilent 34970A, USA) and a computer. Fig. 2 exhibits the experimental facilities of radiation measurement. The left is the structure diagram of the measurement system, and the right is the real objects. The solar-absorbing efficiency or energy conversion efficiency from solar radiation to heat stored into PCM ($\eta_{sol\text{-}pcm}$) can be calculated by the ratio of latent heat stored in the PCM to total solar radiation during one phase change period (Eq. (1)) (Chen et al., 2012).

$$\eta_{sol-pcm} = \frac{m_{pcm}\Delta H_{pcm}}{\int_{t_s}^{t_e} P(t) \times 3600 \times S} \quad (1)$$

where m_{pcm} (g) is the total mass of PCM, ΔH_{pcm} (J/g) is the phase change enthalpy, $P(t)$ (W/m^2) is the solar radiation intensity which is a time-dependent value. S (m^2) is the total irradiated area. In this work, it is equal to 0.0814 m^2. t_s (h) and t_e (h) are the start and end time of one phase change process, respectively. The $\eta_{sol\text{-}pcm}$ is used to assess solar-to-thermal conversion performance of PCM, which a decisive factor restricting thermal storage rate of the system.

Once the PCM finishes the phase change process or reaches the preconcerted temperature (70 °C in our experiment), the pump stalled in the pipeline system is turned on. A heat exchange process between the PCM and water is conducted to evaluate the practical application effects of the water heating system. 10 L water is stored in the water tank in advance, and its initial temperature is 25 °C. The temperature of the PCM and the water in the tank are continuously scanned using Agilent data logger until two temperatures reaching equilibrium.

The energy efficiency from PCM to water ($\eta_{pcm\text{-}water}$) is defined as the ratio of heat absorbed by the water to the heat released by the PCM and the PCM chamber (the sensitive heat of PCM chamber cannot be ignored). As a result, the energy efficiency of solar radiation to heat stored in water ($\eta_{sol\text{-}water}$) or solar-to-thermal conversion efficiency of the system can be defined as the product of $\eta_{sol\text{-}pcm}$ and $\eta_{pcm\text{-}water}$, as shown in Eq. (2). The detailed calculation process can be found in Supporting Information (Eq. S1 to S7).

$$\eta_{sol-water} = \eta_{sol-pcm} \times \eta_{pcm-water} \quad (2)$$

Table 1
Phase change properties of composite PCMs.

Sample	Melting enthalpy (J/g)	Melting temperature (°C)
SAT	271.2	58.7
SAT/GO	267.7	58.7
SAT/EG	218.1	57.9
SAT/GO/EG	215.7	57.7

3. Results and discussion

3.1. Thermophysical properties of composite PCMs

Fig. 3(a) gives the DSC curves of four samples, and detailed values are listed in Table 1. It can be seen that the phase change enthalpy of SAT/GO was equal to 267.7 J/g, very close to the enthalpy of SAT (271.2 J/g). Both the melting temperatures of SAT and SAT/GO were 58.7 °C, which indicates that the GO has little effect on the phase change properties of SAT. The phase change enthalpies of SAT/EG and SAT/GO/EG were 218.1 J/g and 215.7 J/g, respectively, equivalent to their calculated one based on the mass fractions of SAT or SAT/GO in the composites PCMs. It is suggested that the composite PCM can maintained the suitable phase change temperature (about 58 °C) and high enthalpy, making them show great promise in the application of solar water heating system.

Fig. 3(b) shows the thermal weight loss curves of four samples. Four samples started to decompose at the beginning of test procedure and lost all water of crystallization gradually. There were two weight loss stages in each curve, corresponding to the thermal decomposition of $NaAC{\cdot}3H_2O$ and NaAC. The weight loss on the first stage for SAT and SAT/GO were 39.0% and 39.3%, respectively, very close to theatrical mass fraction of water of crystallization (38.9%). For the SAT/EG and SAT/GO/EG, the mass changes were equal to 31.7% and 30.3%, respectively. These values were also proportional to the calculated weight fractions of water in the composite PCM (31.1%).

3.2. Thermal conductivity of composite PCMs

Thermal conductivity can affect the heat exchange efficiency considerably and is influenced by the sample density. Hence, the thermal conductivities of SAT/EG and SAT/GO/EG with various densities were measured. Fig. 4 illustrates that the thermal conductivities of composite PCM almost increased linearly with the increase of PCM densities. It should be noticed that the thermal conductivities of SAT/EG were always higher than those of SAT/GO/EG at the same density. That is to say, adding GO weakens the thermal conductivity of SAT/GO instead of strengthening it, which was not consistent with many researches reported in other works (Liu et al., 2019; Lin et al., 2019). The content of GO in the SAT/GO/EG composite (less than 0.1 wt%) may be responsible for this phenomenon. The large amount of GO can form effective thermal conductive network that promotes the heat transport, while trace amount of GO dispersed in SAT/EG only causes obvious interface thermal resistance between GO and SAT/EG composite (Fan and Khodadadi, 2011; Qureshi et al., 2018).

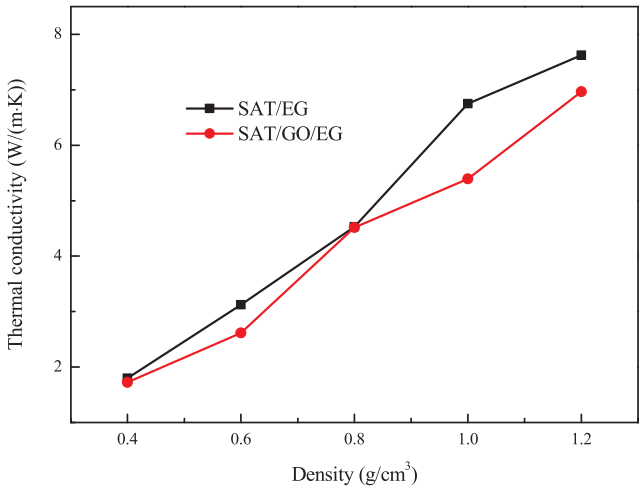


Fig. 4. Thermal conductivities of composite PCMs with different densities.

3.3. Determining the optimum density of SAT/EG and SAT/GO/EG

As described above, the volume of PCM chamber is a fixed value and equal to 1628 cm^3. As a result, the mass of PCM was directly determined by its density. The higher the density is, the greater the mass is. High density cannot only be in favour of desirable thermal conductivity, but help to improve the heat storage capacity. However, compressed PCM with high density may confront lower shape stability, i.e. leakage. Consequently, the optimum density of composite PCM should be determined in order to improve thermal storage capacity of the water heating system as well as maintain the shape stability of PCMs as much as possible.

The loose PCMs were compressed into blocks with various densities (0.4 g/cm^3, 0.6 g/cm^3, 0.8 g/cm^3, 1.0 g/cm^3, 1.2 g/cm^3). These blocks were placed in seal beakers and heated to 80 °C for 2 h. After the PCMs cooling to room temperature, a careful observation was conducted to the beakers.

As displayed in the Fig. 5(a), no any residuals are found in three breakers for placing the composite SAT/EG at the densities of 0.4 g/cm^3, 0.6 g/cm^3 and 0.8 g/cm^3, while some white residuals of SAT can be clearly observed in the breakers when the density is 1.0 g/cm^3 or 1.2 g/cm^3. These results indicate that the density of SAT/EG used in this work should be less than 1.0 g/cm^3. However, it can be observed that there was no residual on the breaker when the density is 1.0 g/cm^3 for the composite SAT/GO/EG. In order words, the compressed SAT/GO/EG at this density can maintain nice shape stability. Namely, the shape stability of SAT/GO/EG was superior to that of SAT/EG. GO is hydrophilic and contains abundant oxygen-containing functional groups, such as hydroxyl, carboxyl and epoxy group (Al-Gaashani et al., 2019; Qureshi et al., 2019). The presence of these functional groups is the main reason for improved shape stability. Therefore, the most suitable density of PCMs applied in the water heating system should be 0.8 g/cm^3 for SAT/EG and 1.0 g/cm^3 for SAT/GO/EG.

3.4. The influence of GO on solar-absorbing performance

Many researchers applied carbon nanomaterials to boost solar-absorbing performance of PCM (Wang et al., 2013; Yang et al., 2016). Therefore, in our work, the addition of GO also intended to increase sunlight absorption of composite PCM. The SAT/EG and SAT/GO/EG blocks at the same density (0.8 g/cm^3) were directly exposed to the sun. The temperature variation curves of two samples and solar radiation intensity are given in Fig. 6.

Under solar radiation, the temperature of SAT/EG and SAT/GO/EG sharply increased from 33.8 °C to about 57 °C. This process belongs to sensible heat storage stage. After that, an inflection point along with a temperature platform appeared which corresponded to the phase change stage. The phase change time spent by the SAT/EG was very close to that spent by the SAT/GO/EG. Therefore, It can be concluded that the addition of GO cannot virtually contribute to the solar absorption and solar-to-thermal conversion. After the phase change processes completed, two PCMs undertook another temperature rise, and heating rate of SAT/EG was slightly higher than that of SAT/GO/EG. The peak temperature of SAT/EG was about 80.3 °C while it was only 77.2 °C for SAT/GO/EG. Similar to the thermal conductivity, the solar-absorbing performance of SAT/EG cannot be elevated by adding minute

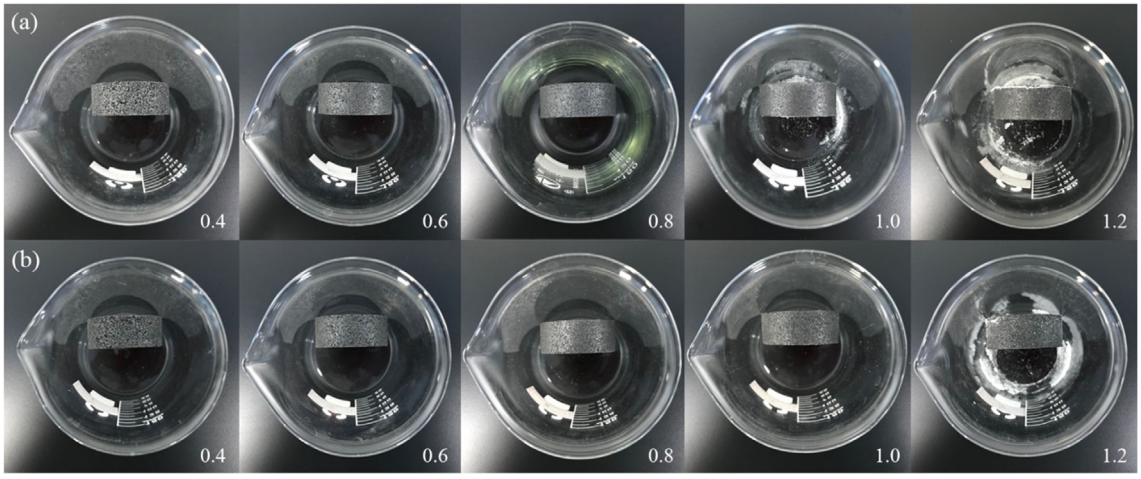


Fig. 5. Photographs of PCM blocks with different densities after being heated at 80 °C: SAT/EG (a) and SAT/GO/EG (b).

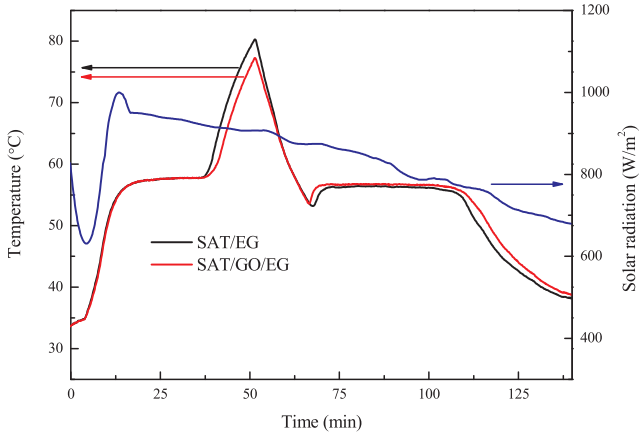


Fig. 6. Temperature variation of SAT/EG and SAT/GO/EG under solar radiation (1:15 PM to 2:08 PM, 26 July 2019, Zhuhai, China).

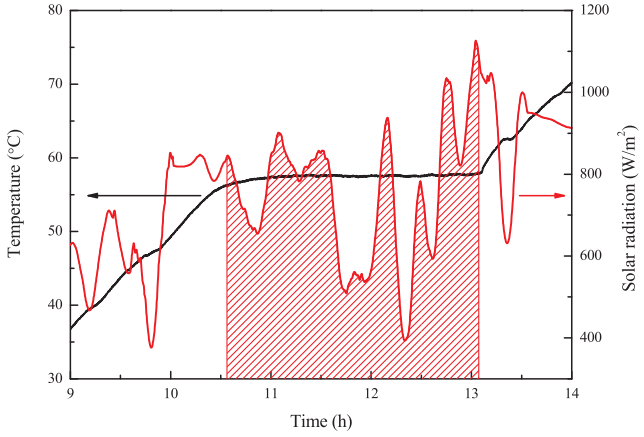


Fig. 7. Temperature evaluation of SAT/GO/EG under solar radiation from 9:00 AM to 2:00 PM (26 July 2019, Zhuhai, China).

amount of GO. Inversely, it was slightly weakened because of lower thermal conductivity of SAT/GO/EG compared with that of SAT/EG.

The solar radiation lasted for 51.4 min, and then the samples were sheltered. Two temperature curves showed a sharp downward trend. The supercooling temperature was less than 3.3 °C for both two kinds of PCMs, indicating that the composite PCMs were characterized by good crystallization property.

3.5. Performance evaluation of water heating system

Though GO failed to accelerate light absorption, nevertheless, it increased the shape stability of composite PCMs, as discussed in Section 3.3. The thermal storage capacity of system is thus enlarged using the SAT/GO/EG. Therefore, SAT/GO/EG in the density of 1.0 g/cm^3 was employed to build the water heating system.

3.5.1. Solar energy storage

Fig. 7 gives the temperature rise curves of SAT/GO/EG installed in the system and corresponding solar radiation intensity (red curve). The shaded area between radiation intensity and x-axle represents the integral of solar intensity for time at the range of phase change process of PCMs, i.e. $\int_{t_s}^{t_e} P(t)$ in Eq. (1). It can be obtained using the soft of Origin. Table 2 lists detailed parameters in Eq.1.

The solar-absorbing efficiency was calculated to be 63.7%, as shown in Table 2. That is to say, 63.7% of solar energy is absorbed by the PCM and transferred to the thermal energy, which exceeds the most reports related to solar water heating system (Jouybari et al., 2019; Maraj et al.,

Table 2
The parameters used to calculate the solar-absorbing efficiency ($\eta_{sol\text{-}pcm}$).

	m (g)	ΔH_m (J/g)	$\int_{t_s}^{t_e} P(t)$(Wh/m^2)	S (m^2)	$\eta_{sol\text{-}pcm}$ (%)
SAT/GO/EG	1628.0	215.7	1880	0.0814	63.7

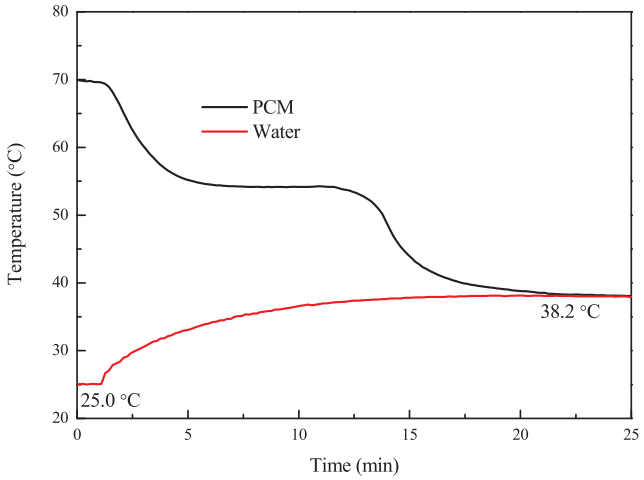


Fig. 8. Temperature evolution of SAT/GO/EG and 10 L water in the heat exchange process.

2019; Pranesh et al., 2019).

3.5.2. Solar energy release

A heat exchange process was conducted to verify the application effects of water heating system. Fig. 8 exhibits the temperature evolution of the PCM and water stored in water tank. The temperature of PCM decreased from 70 ˚C to 38.2 ˚C, while the water temperature increased by 13.2 ˚C within 25 min. Using Eqs. (2) and (S1)–(S7), the system efficiency from solar radiation into heat stored into water was calculated to be 54.5%.

In addition, the final water temperature was equal to 38.2 ˚C that is in the comfort temperature level of human bath (35–40 ˚C) (Yang et al., 2016; Kudo et al., 2019). Therefore, our newly designed system will show great promise in the preparation of domestic hot water. Compared with the traditional system, this novel system has higher thermal storage capacity along with wider geographical application.

4. Conclusions

Two composite PCMs of SAT/EG and SAT/GO/EG were prepared in this article. Their thermophysical characteristic and solar-absorbing performance were investigated. Test results indicated that GO showed little effect on the thermal properties and solar absorption performance of composite PCM. However, it can significantly improve the shape stability of composite PCM. The higher the density is, the larger the volumetric heat storage capacity. When the density increased to 1 g/cm^3, SAT/EG showed severe leakage while SAT/GO/EG can still keep the shape stability. A novel solar water heating system was designed using SAT/GO/EG (1 g/cm^3) as the solar-absorbing substance and thermal storage media simultaneously. Under the real solar radiation, the PCM gave a high solar-absorbing efficiency of 63.7%. During a heat exchange process, the temperature of 10 L water can increase from 25 ˚C to 38.2 ˚C within 25 min. The energy conversion efficiency from solar radiation into heat absorbed by water is as high as 54.5%, which indicates that the novel system exhibits great application effects, and the composite PCM of SAT/GO/EG is very promising in designing this novel water heating system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.solener.2020.02.054.

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